K-MEANS CLUSTERING AND PRINCIPAL COMPONENT ANALYSIS

Implementation from Scratch Using the Star Dataset

Dataset	Star Dataset (240 observations, 7 attributes)
Clustering features	Temperature, Luminosity, Radius, Absolute Magnitude
Algorithms	K-Means from scratch and PCA using covariance/eigen decomposition
Experiments	K = 2, 3, 4, 5, 6; PCA with 2 and 3 components

Prepared from the execution results and visualizations generated by the implementation.

Abstract

This assignment implements K-Means clustering and Principal Component Analysis (PCA) on the Kaggle Star Dataset. K-Means is implemented manually using random observation-based centroid initialization, Euclidean distance, nearest-centroid assignment, centroid updates, and Within-Cluster Sum of Squares (WCSS). Experiments are performed for K values from 2 to 6. PCA is also implemented manually by standardizing and centering the feature matrix, calculating its covariance matrix, and applying eigenvalue decomposition. The experiment obtains a four-component representation and projects the observations into two- and three-dimensional spaces. On the reported run, K=4 gives the highest Silhouette score among the tested values, while K=3 gives the highest Calinski-Harabasz score and K=6 gives the lowest Davies-Bouldin index. The first two principal components explain 83.86% of the total variance, while the first three explain 93.19%.

1. Introduction

Clustering is an unsupervised learning task in which observations are grouped according to similarity in their feature values. K-Means partitions observations into K clusters by repeatedly assigning each observation to its nearest centroid and updating each centroid to the mean of the observations assigned to it. WCSS measures the compactness of the resulting clusters. The Elbow Method examines how WCSS changes as K increases to help identify a useful number of clusters.

PCA is a dimensionality-reduction technique that transforms correlated features into a smaller set of orthogonal principal components. The components are ordered by the amount of variance they explain. In this assignment, PCA is used to reduce the four-dimensional standardized star-feature space to two and three dimensions and to visualize the data.

2. Objectives

• Implement K-Means without using a pre-built clustering estimator.
• Test K = 2, 3, 4, 5 and 6 and calculate WCSS for each value.
• Plot the Elbow curve and examine clustering evaluation metrics.
• Implement PCA using a covariance matrix and eigenvalue decomposition.
• Calculate eigenvalues, eigenvectors, explained variance ratios and cumulative explained variance.
• Project the dataset into two and three principal components and visualize the first two components.

3. Dataset Description

The experiment uses the Kaggle Star Dataset. The loaded CSV contains 240 observations and 7 columns. The dataset includes four numerical physical measurements and three descriptive/class columns.

Attribute	Description / use
Temperature (K)	Numerical feature; used as a clustering and PCA input.
Luminosity(L/Lo)	Numerical feature; used as a clustering and PCA input.
Radius(R/Ro)	Numerical feature; used as a clustering and PCA input.
Absolute magnitude(Mv)	Numerical feature; used as a clustering and PCA input.
Star type	Categorical class label encoded as integers; excluded from model input and used only for post-clustering interpretation.
Star color	Categorical descriptor; not included in the numerical model input.
Spectral Class	Categorical descriptor; not included in the numerical model input.

All four selected numerical columns contained 240 non-missing values in the displayed dataset inspection. The clustering input therefore had shape 240 × 4. Star type was deliberately excluded from the feature matrix to avoid using the known class label to form the clusters.

4. Methodology

4.1 Preprocessing and standardization

The four numerical features have very different scales. For example, luminosity ranges from very small decimal values to hundreds of thousands, while absolute magnitude has a much smaller numerical range. K-Means uses Euclidean distance, so features with large numerical scales could dominate the distance calculation. StandardScaler was therefore applied to each feature. For a feature x, standardization is given by $z = (x - \mu) / \sigma$, where μ is the feature mean and σ is its standard deviation.

4.2 K-Means implementation

The K-Means algorithm was implemented using NumPy rather than a pre-built K-Means estimator. The steps are:

1. Select K distinct observations randomly as the initial centroids.
2. Calculate the Euclidean distance from each observation to every centroid.
3. Assign each observation to the cluster with the nearest centroid.
4. Update each centroid by calculating the mean of all observations assigned to that cluster.
5. Repeat assignment and update until centroid movement is below the tolerance or the maximum iteration count is reached.
6. Calculate WCSS for the resulting clusters.

The Euclidean distance between observation x_i and centroid μ_j is $d(x_i, \mu_j) = \sqrt{\sum_j (x_{ij} - \mu_{ij})^2}$. WCSS is the sum of squared distances from each observation to its assigned centroid: $WCSS = \sum_j \sum_{i \in C_j} ||x_i - \mu_j||^2$. To reduce sensitivity to a single random initialization, the implementation performs 10 initializations for each K and retains the run with the lowest WCSS. Empty clusters are reinitialized to a randomly selected observation.

4.3 Selection and evaluation of K

The Elbow Method plots WCSS against K. WCSS generally decreases as K increases because more centroids allow observations to be represented more closely. The decrease must therefore be interpreted alongside cluster-quality measures. Three additional metrics were calculated: Silhouette score (higher is generally better), Davies-Bouldin index (lower is generally better), and Calinski-Harabasz score (higher is generally better).

4.4 PCA implementation

PCA was implemented from the standardized feature matrix. First, the feature-wise mean was subtracted to center the data. The covariance matrix was calculated using the centered observations. Eigenvalue decomposition was then performed on this symmetric covariance matrix. Eigenvalues and their corresponding eigenvectors were sorted in descending order of eigenvalue. The explained variance ratio for component i is $\lambda_i / \sum_i \lambda_i$, and the cumulative explained variance is the running sum of these ratios. Finally, the centered observations were projected onto the first two and first three eigenvectors.

5. Results and Discussion

5.1 K-Means experimental results

The following results were printed by the executed program for the 240 observations. WCSS decreases as K increases, as expected. The table also reports the number of iterations and three cluster evaluation metrics.

K	WCSS	Iterations	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	507.7393	7	0.504553	0.972969	211.994685
3	332.8041	14	0.529402	0.977085	223.322746
4	256.8954	13	0.536674	0.943765	215.305137
5	213.9239	12	0.463212	1.025012	204.895191
6	169.3319	16	0.506682	0.870733	218.525123

The Silhouette score is highest at K=4 (0.536674), indicating the strongest average separation/cohesion among the tested values according to this measure. The Calinski-Harabasz score is highest at K=3 (223.322746). The Davies-Bouldin index is lowest at K=6 (0.870733). Since these measures do not identify the same K, the choice should not be described as an absolute optimum. Considering the Elbow curve together with the best Silhouette score and the Davies-Bouldin value at K=4, K=4 is used as a practical representative choice for the final PCA cluster visualization.

5.2 Elbow curve

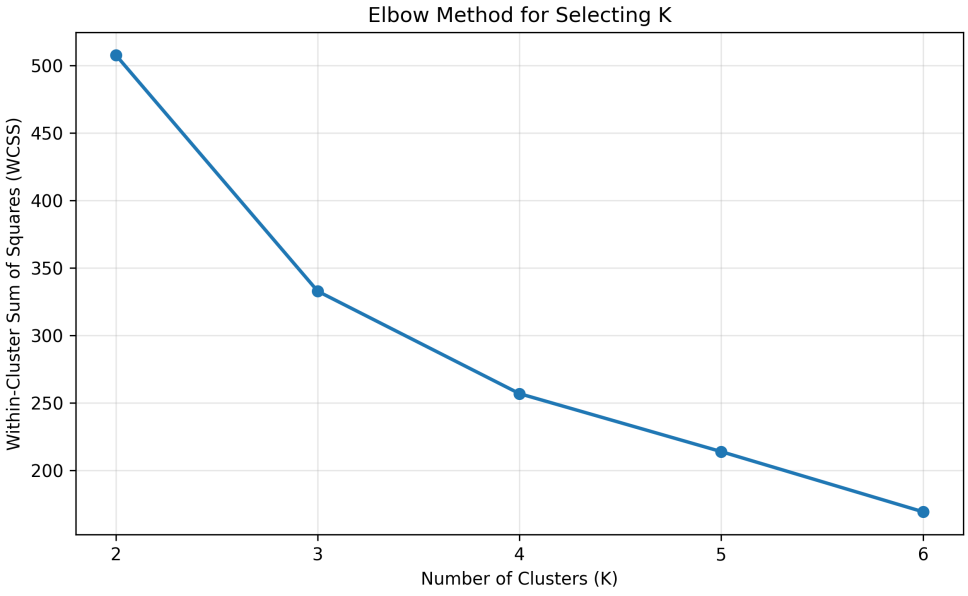


Figure 1. Elbow curve showing WCSS for K = 2 to K = 6.

The WCSS reduction is large when moving from K=2 to K=3, and the curve begins to flatten as K increases further. This suggests that the useful region is around K=3 to K=4, although the elbow is not perfectly sharp. The curve should be interpreted together with the evaluation metrics rather than used alone.

5.3 Convergence and cluster distribution

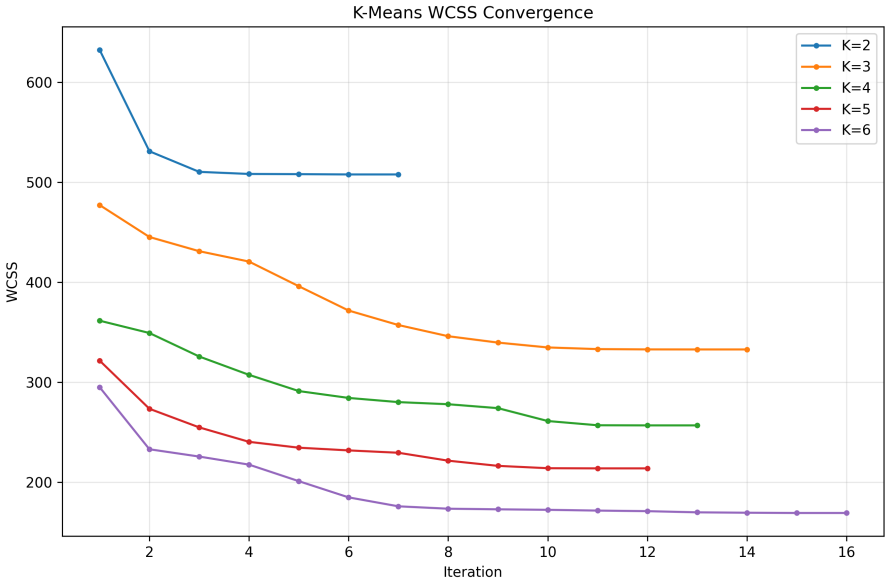


Figure 2. WCSS convergence across iterations for the tested K values.

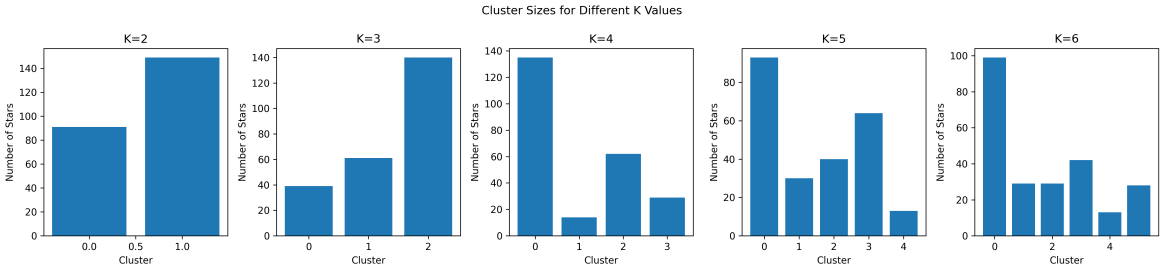


Figure 3. Number of observations assigned to each cluster for each K.

The convergence plot shows the reduction of WCSS as centroids are updated. The cluster-size plot provides a view of how the 240 observations are distributed across the resulting clusters. Cluster IDs are arbitrary identifiers; a cluster number does not represent an inherent star category.

5.4 Evaluation metric plots

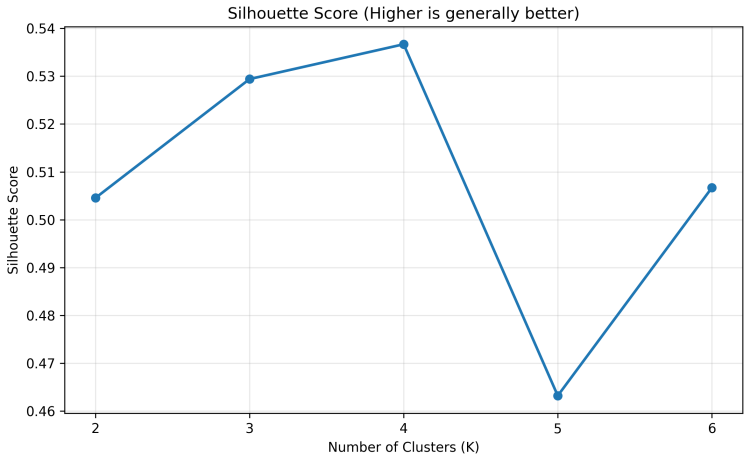


Figure 4. Silhouette score by K.

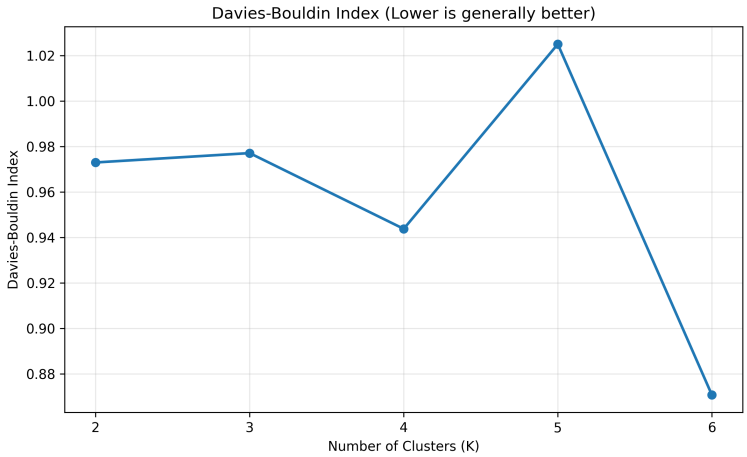


Figure 5. Davies-Bouldin index by K.

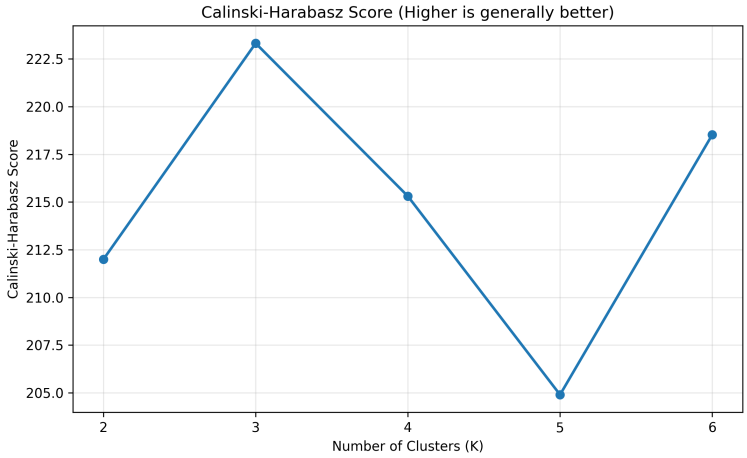


Figure 6. Calinski-Harabasz score by K.

6. PCA Results

6.1 Covariance matrix and eigenvalue decomposition

The covariance matrix was calculated from the standardized, centered data. Eigenvalue decomposition produced four eigenvalues, corresponding to the four input features. The eigenvalues, sorted in descending order, were 2.42439729, 0.94396455, 0.37474725 and 0.27362730. The corresponding eigenvectors define the directions of the principal components.

6.2 Explained variance

Component	Eigenvalue	Explained variance	Cumulative variance
PC1	2.424397	60.36%	60.36%
PC2	0.943965	23.50%	83.86%
PC3	0.374747	9.33%	93.19%
PC4	0.273627	6.81%	100.00%

The first principal component explains 60.36% of the total variance. The first two components together explain 83.86%, and the first three explain 93.19%. Thus, reducing the four-dimensional input to two dimensions retains most of the variance and provides a useful visualization, while three components retain more than 90% of the variance.

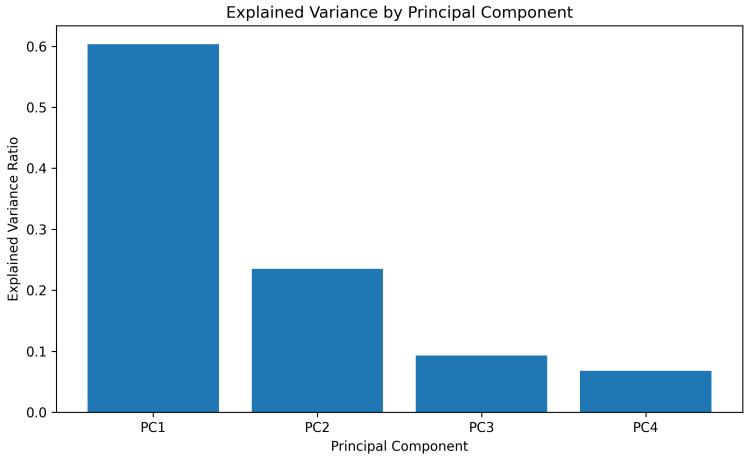


Figure 7. Explained variance ratio for each principal component.

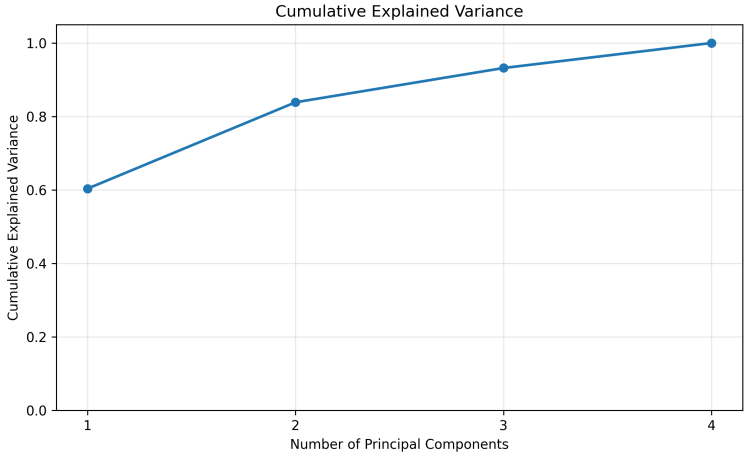


Figure 8. Cumulative explained variance as components are added.

6.3 Two-dimensional PCA projection

Each standardized observation was projected onto the first two eigenvectors. PC1 accounts for 60.36% of the variance and PC2 accounts for 23.50%. The scatter plot displays the 240 stars in the resulting two-dimensional representation. PCA does not use Star type labels when computing the components.

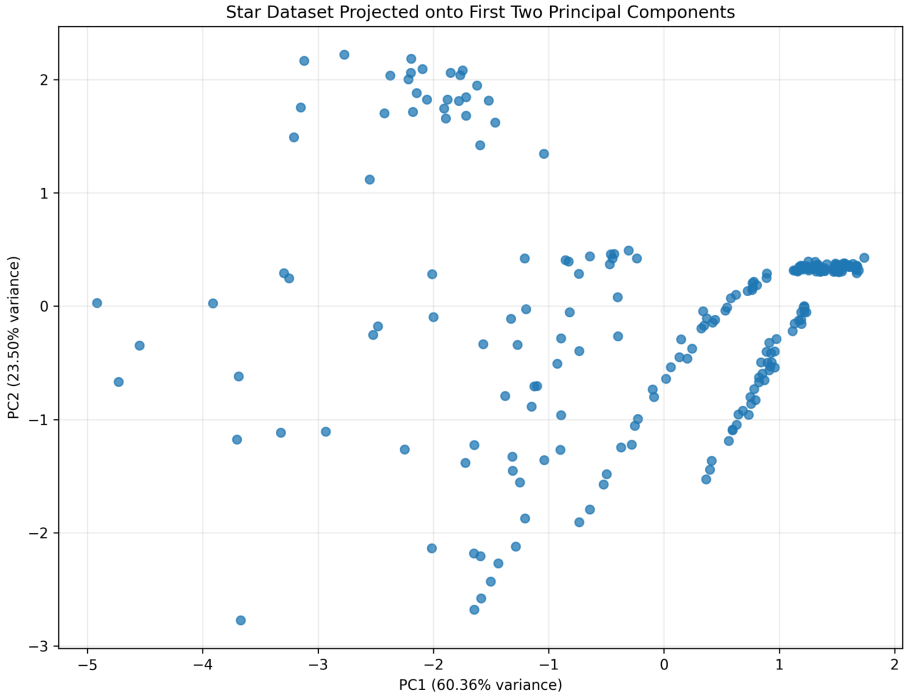


Figure 9. All observations projected onto PC1 and PC2 (the first two components explain 83.86% of variance).

6.4 K-Means clusters projected into PCA space

For the final cluster visualization, K-Means labels obtained in the original four-dimensional standardized feature space for K=4 were displayed using the first two PCA coordinates. This is a projection of the original clustering result, not a new K-Means fit in two dimensions. As a result, points that appear close or overlap in the two-dimensional plot may still be separated in the original four-dimensional space.

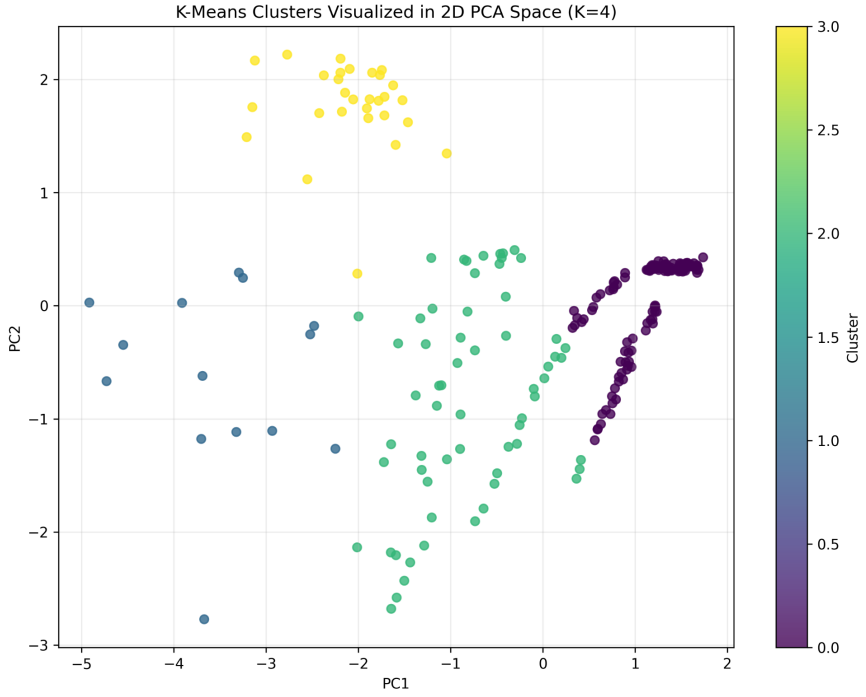


Figure 10. Original four-dimensional K-Means assignments (K=4) visualized in the first two PCA dimensions.

6.5 Three-component representation

The implementation also generated a three-component transformed dataset. The first three principal components explain 93.19% of the total variance. This representation retains more information than the two-component representation, at the cost of requiring a three-dimensional visualization.

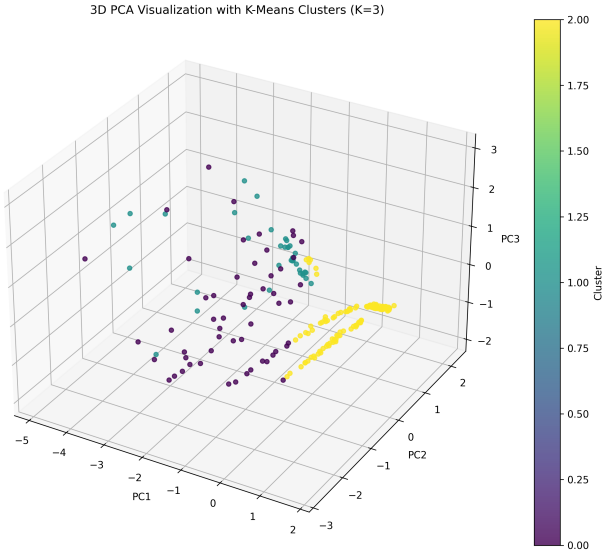


Figure 11. Three-dimensional PCA visualization with cluster coloring (generated by the implementation).

7. Discussion

The experiment demonstrates the effect of increasing K on cluster compactness. WCSS decreases monotonically from 507.7393 at K=2 to 169.3319 at K=6. This is expected because increasing the number of centroids gives the algorithm more flexibility to reduce within-cluster distances. The Elbow curve suggests diminishing returns around K=3–4.

The evaluation metrics provide complementary perspectives. K=4 has the highest Silhouette score, suggesting the strongest balance of within-cluster cohesion and separation among the tested values. K=6 has the lowest Davies-Bouldin index, while K=3 has the highest Calinski-Harabasz score. The disagreement reinforces that cluster selection involves a trade-off and should consider the curve, metrics, cluster sizes and the purpose of the analysis. K=4 was selected for the final visualization as a reasonable compromise, not as a universally proven optimum.

PCA shows that the four original standardized features can be represented with relatively little information loss in a lower-dimensional space. Two components retain 83.86% of the variance, and three retain 93.19%. The PCA plots make the structure easier to inspect visually. However, a two-dimensional projection cannot preserve every distance and separation present in the original four-dimensional feature space.

8. Limitations

• K-Means is sensitive to feature scaling and initial centroid selection; standardization and multiple initializations reduce, but do not eliminate, this sensitivity.
• K-Means favors compact, approximately spherical clusters and may not represent every possible cluster shape.
• The dataset is relatively small (240 observations), so results describe this dataset and should not automatically be generalized to all stars.
• PCA preserves directions of maximum variance, not necessarily the dimensions most useful for separating star categories.
• The evaluation metrics disagree about the best K, so the selected K should be justified as a practical choice rather than a definitive ground truth.

9. Conclusion

K-Means clustering and PCA were successfully implemented on the Star Dataset. The K-Means implementation tested K values from 2 to 6 and calculated WCSS for each value. The results showed decreasing WCSS with increasing K, while the Elbow curve indicated diminishing improvement around K=3–4. Considering the evaluation metrics together, K=4 was used for the final cluster visualization, although K=3 and K=6 performed best on individual alternative metrics. PCA was implemented using the covariance matrix and eigenvalue decomposition. The first two principal components explained 83.86% of the variance, and the first three explained 93.19%. The resulting two-dimensional and three-dimensional representations provide useful visual summaries of the dataset.

10. Reproducibility and Files

The Python implementation is provided separately as **star_clustering_pca.py**, together with **requirements.txt** and a README file. The dataset CSV must be placed at the path configured in the script. The program saves plots and result tables in an **outputs** directory. The reported values in this document are taken from the execution output supplied for this run.

To run the implementation:

```
pip install -r requirements.txt
python star_clustering_pca.py
```

References

1. Kaggle, “Star Dataset,” Deepu1109. https://www.kaggle.com/datasets/deepu1109/star-dataset
2. Lloyd, S. (1982). “Least squares quantization in PCM.” IEEE Transactions on Information Theory, 28(2), 129–137.
3. Jolliffe, I. T., & Cadima, J. (2016). “Principal component analysis: a review and recent developments.” Philosophical Transactions of the Royal Society A, 374(2065).